Neuroscience-Inspired Cognitive Architecture for Multi-Agent Systems: Implementing Human-Like Cognition Through the ORPA Framework

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Abstract

Current multi-agent systems predominantly operate as sophisticated workflow automation tools, lacking the cognitive depth that characterizes human expert reasoning. This paper presents a novel cognitive architecture for multi-agent systems grounded in established neuroscience principles, specifically predictive coding theory, dual memory systems, selective attention mechanisms, and metacognitive processes. We introduce the ORPA (Observe, Reflect, Plan, Act) cognitive framework that translates core neuroscientific findings into implementable multi-agent architectures. Our approach addresses fundamental limitations in current agent designs by incorporating prediction error minimization, episodic-semantic memory integration, attention-based information filtering, and metacognitive self-monitoring. Through analysis of the XMPro Multi-Agent Generative System (MAGS) implementation and comparison with Stanford's Generative Agents research, we demonstrate that neuroscience-grounded cognitive architectures can produce more adaptive, reliable, and human-like agent behavior. We further show that the same four stages compute the elements of Warren Powell's universal model for sequential decisions, so that ORPA is, in decision-theoretic terms, a hybrid policy of the kind that decision science finds works in practice. Situating the language model as exogenous information inside a deterministic policy function makes the governing decision reproducible and auditable, and it lets the agent learn from operating outcomes through closed-form Bayesian updating without relaxing its safety constraints. The neuroscience and decision-science groundings reinforce one another, most directly in the surprise signal, which is at once a prediction error and a quantity of Shannon information. This work contributes to the emerging field of cognitive multi-agent systems by providing both theoretical foundations and practical implementation strategies for creating agents that genuinely emulate human expert cognition rather than merely automating predefined tasks.

Keywords: multi-agent systems, cognitive architecture, neuroscience, predictive coding, metacognition, industrial AI, sequential decision analytics, policy, belief state

1 Introduction

The field of multi-agent systems has achieved remarkable advances in computational efficiency and task automation, yet most contemporary systems fundamentally operate as sophisticated workflow orchestrators rather than cognitive entities capable of genuine reasoning and adaptation [1]. While these systems excel at executing predefined sequences and processing large volumes of data, they lack the cognitive sophistication that enables human experts to observe patterns, reflect on experience, and adapt strategies based on accumulated understanding.

This cognitive limitation becomes particularly evident in complex, dynamic environments where rigid rule-based approaches prove insufficient. Industrial operations require the nuanced decision-making capabilities that distinguish experienced operators from novices: the ability to detect subtle anomalies, integrate contextual knowledge, and make uncertainty-aware decisions under changing conditions [2].

Recent breakthroughs in generative AI and large language models (LLMs) have created new opportunities to address this cognitive gap. The Stanford Generative Agents research demonstrated that structured cognitive architectures built on LLM foundations can produce remarkably human-like behavior through systematic implementation of memory, reflection, and planning mechanisms [3]. However, translation of these research insights into practical industrial applications remains limited.

This paper presents a comprehensive framework for implementing neuroscience-inspired cognitive architecture in multi-agent systems, specifically through the ORPA (Observe, Reflect, Plan, Act) cognitive cycle.

The framework has an intellectual lineage worth stating, because each layer answers a question the previous one could not. Boyd's OODA loop contributed the continuous

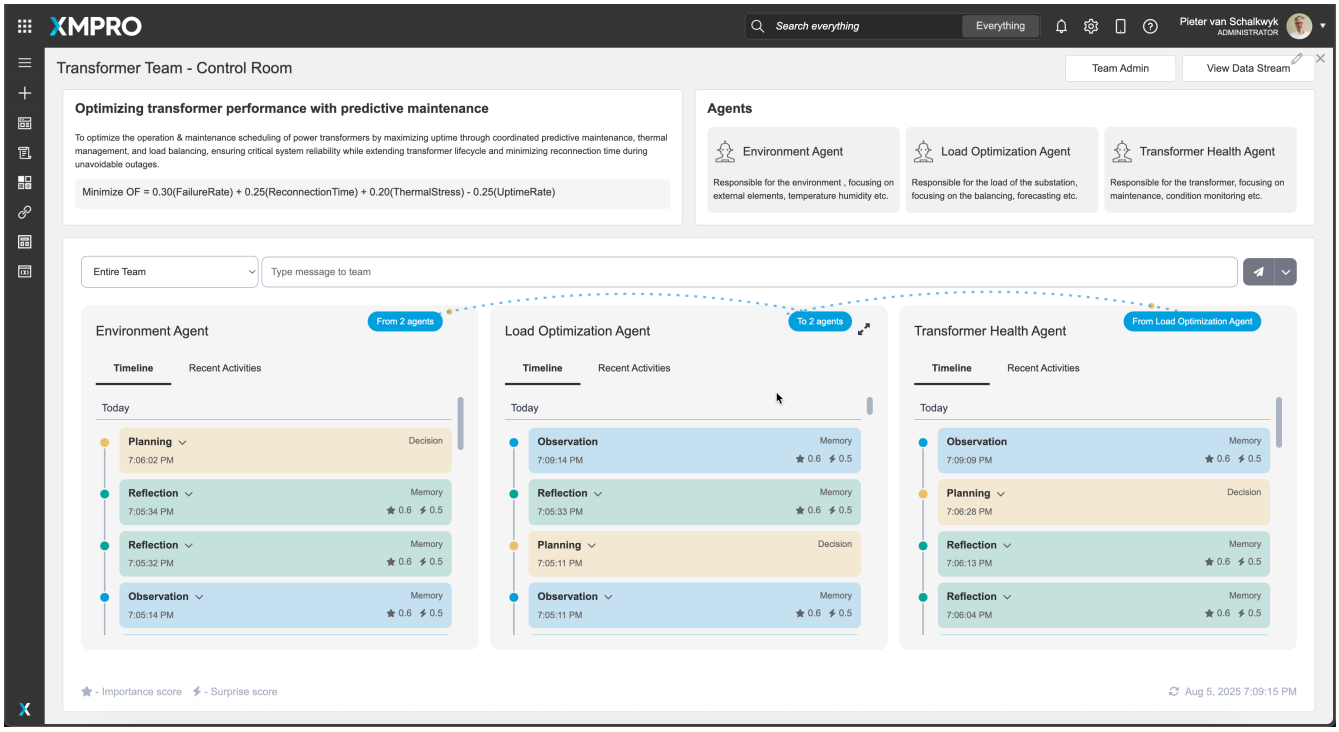


Figure 1: Multi-Agent Generative System (MAGS) demonstrating a team of three cognitive agents, each implementing their own ORPA (Observe, Reflect, Plan, Act) cycle. The Environment Agent, Load Optimization Agent, and Transformer Health Agent work collaboratively towards the team's Objective Function of optimizing transformer performance with predictive maintenance, resolving differences through consensus mechanisms while maintaining individual cognitive autonomy.

decision cycle, Park et al.'s Generative Agents contributed the cognitive architecture of memory and reflection, and the industrial implementation discussed here hardened that architecture for operations. Powell's Sequential Decision Analytics supplies the fourth layer, namely the decision science that specifies what each stage of the cycle should compute and how to demonstrate that it computes it correctly. The first three layers are already present in the cognitive account below; this paper adds the fourth.

Our contributions include:

1. A systematic mapping of established neuroscience principles to multi-agent system design
2. The ORPA cognitive framework for implementing human-like reasoning in artificial agents
3. Practical implementation strategies demonstrated through the XMPro MAGS platform
4. Analysis of cognitive enhancement capabilities and deployment considerations
5. A decision-theoretic formalization of ORPA in the language of Sequential Decision Analytics, including the mapping of its stages to a universal decision model, a determinism result that situates language-model stochasticity as bounded exogenous information, and a bounded-learning mechanism in which beliefs and policy parameters are learnable while the constraint set is not.

2 Related Work

2.1 Cognitive Architectures in AI

The development of cognitive architectures has been a central pursuit in artificial intelligence since its inception. Early work by Newell and Simon established the foundation for symbolic cognitive architectures such as ACT-R [4] and SOAR [5], which attempt to model human cognition through rule-based reasoning systems. However, these approaches have shown limitations in handling uncertain, dynamic environments that characterize real-world applications.

More recent work has focused on integrating neural and symbolic approaches. The Global Workspace Theory [6] has influenced architectures that attempt to model conscious awareness and attention mechanisms. However, these systems typically require extensive domain-specific engineering and struggle with the flexibility demands of multi-agent environments.

2.2 Generative Agents and Emergent Behavior

A significant breakthrough in cognitive agent research emerged from Park et al.'s Generative Agents study [3], which demonstrated that LLM-based agents with structured memory and reflection mechanisms could produce believable

human-like behavior in social simulations. The key innovation was not the underlying language model capability, but rather the cognitive architecture that enabled agents to:

- Store and retrieve episodic memories of specific experiences
- Reflect on accumulated experiences to generate higher-level insights
- Plan future actions based on these integrated understandings
- Maintain behavioral consistency across extended interactions

This work provided empirical evidence that carefully designed cognitive architectures could bridge the gap between computational capability and genuine intelligence-like behavior.

2.3 Neuroscience-Informed AI Systems

The integration of neuroscience principles into AI system design has gained increasing attention as our understanding of brain function deepens. Predictive coding theory, originally proposed by Rao and Ballard [7] and extensively developed by Friston [8], suggests that the brain operates as a hierarchical prediction machine that minimizes surprise through active inference.

Recent work has begun exploring how predictive coding principles can inform AI architecture design. Clark [9] argues that predictive processing represents a unifying framework for understanding biological intelligence that could guide artificial system development. However, practical implementations of these principles in multi-agent systems remain limited.

2.4 Industrial Multi-Agent Systems

Current industrial multi-agent systems focus primarily on workflow orchestration and distributed task execution [10]. These systems excel at coordinating multiple specialized agents to accomplish complex tasks but typically lack the cognitive sophistication required for adaptive reasoning in dynamic environments.

The limitation of these approaches becomes apparent in scenarios requiring expert-level judgment, such as anomaly detection in complex systems, process optimization under changing conditions, or safety-critical decision making with incomplete information [11]. These situations demand cognitive capabilities that distinguish human experts from automated systems.

2.5 Sequential Decision Analytics

Decisions under uncertainty are studied by many distinct communities, among them stochastic programming, Markov decision processes, reinforcement learning, optimal control, and multi-armed bandits, each with its own notation and preferred solution method. Powell's central observation, developed over four decades and consolidated in *Reinforcement Learning and Stochastic Optimization* [28] and *Sequential Decision Analytics and Modeling* [29], is that these communities address the same underlying problem, and that their apparent differences are differences of policy class rather than of problem. Sequential Decision Analytics (SDA) writes any such problem with five elements, namely a state variable, a decision variable, exogenous information arriving between decisions, a transition function, and an objective defined over policies rather than over individual decisions. The discipline it prescribes is to model first and then solve, and to tune a policy against cumulative performance rather than single-decision quality.

Two features of SDA matter for the present work. First, the object a system executes is not a decision but a policy, a function that maps state to decision and that can be parameterized and tuned. Second, every policy anyone has proposed falls into one of four classes or a hybrid of them: policy function approximations, cost function approximations, value function approximations, and direct lookahead approximations. Powell's practical finding is that policy function approximations, cost function approximations, and deterministic direct lookaheads are by far the most widely used in working systems, whereas value function approximations and the reinforcement learning built on them dominate publications more than plants. This finding identifies, in advance, the industrially robust region of the design space, and Section 5 shows that ORPA already occupies it. The framework also gives learning a precise location: Powell decomposes the state into a physical or resource component, an information component, and a belief component, the last being probability distributions over quantities that cannot be observed directly, and it is the belief component that makes a system capable of learning [28, 30].

3 Neuroscientific Foundations

3.1 Predictive Coding and Active Inference

The Predictive Coding framework represents one of the most influential contemporary theories of brain function [8]. This framework posits that the brain operates as a hierarchical generative model that constantly predicts incoming sensory information and updates its internal representations based on prediction errors.

Core Mechanisms:

- **Top-down Predictions:** Higher cortical levels generate predictions about lower-level sensory input
- **Prediction Error Computation:** Actual sensory input is compared against predictions
- **Hierarchical Error Propagation:** Significant prediction errors propagate up the cortical hierarchy
- **Model Updating:** Persistent prediction errors trigger updates to internal generative models

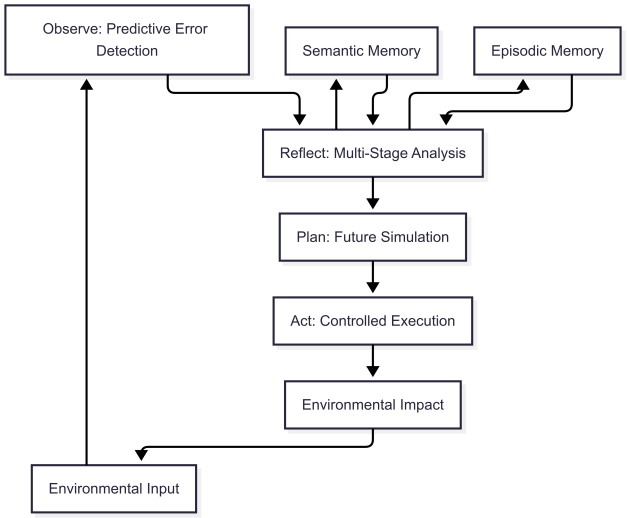


Figure 2: Mapping of neuroscience principles to ORPA cognitive framework components. Predictive coding drives surprise-based attention in the Observe stage, dual memory systems enable episodic-semantic integration during Reflect, and metacognition provides confidence calibration throughout the process.

This mechanism explains how biological systems achieve efficient perception and learning through surprise minimization rather than passive information processing [12]. Organisms actively seek to minimize surprise (technically, the negative log probability) of their sensory observations by either updating their internal models or taking action to change their environment.

Implications for Agent Design: Traditional agent observation mechanisms that passively record and classify events fundamentally misalign with biological perception principles. A neuroscience-grounded approach would implement:

- Continuous expectation generation based on learned environmental models
- Active prediction error detection and prioritization
- Surprise-based attention allocation rather than importance-based filtering

3.2 Dual Memory Systems: Episodic and Semantic Integration

Extensive neuroscience research has established that human memory operates through distinct but interconnected systems [13]. The episodic-semantic distinction, originally proposed by Tulving [14], remains fundamental to understanding how biological systems integrate specific experiences with general knowledge.

Episodic Memory System:

- Stores specific, personally experienced events with rich contextual detail
- Enables "mental time travel" through autonoetic consciousness
- Provides the raw experiential data for reflection and learning
- Associated with hippocampal-neocortical memory networks [15]

Semantic Memory System:

- Contains abstract, context-free knowledge about the world
- Represents generalized patterns, rules, and conceptual relationships
- Forms through the gradual extraction of regularities from episodic experiences
- Associated with distributed neocortical representations [16]

Memory Integration Process: Human reflection involves the dynamic interaction between these systems, where specific episodic memories are retrieved and used to validate, challenge, or update general semantic knowledge. This process, termed "semanticization," is how individual experiences become lasting wisdom [17].

Agent Architecture Implications: Effective agent reflection requires architectural separation of raw experiential data (episodic-like memory) from extracted knowledge representations (semantic-like memory), with explicit mechanisms for integrating specific experiences with general understanding.

3.3 Selective Attention and Cognitive Control

Biological attention systems manage the fundamental challenge of limited cognitive resources in information-rich environments through sophisticated control mechanisms [18]. Research has identified two primary attention control systems.

Top-Down (Voluntary) Control:

- Goal-directed attention based on current behavioral objectives
- Mediated by prefrontal cortex regions that bias processing toward task-relevant information
- Enables sustained focus on planned activities despite environmental distractions [19]

Bottom-Up (Stimulus-Driven) Control:

- Automatic capture by salient or unexpected environmental events
- Ensures sensitivity to potentially important changes that might require behavioral adaptation
- Mediated by ventral attention networks that interrupt ongoing processing [20]

The interaction between these systems creates adaptive attention allocation that balances planned goal pursuit with environmental responsiveness, which is critical for effective agent behavior in dynamic environments.

3.4 Metacognition and Self-Monitoring

Metacognition represents the ability to monitor and evaluate one's own cognitive processes, constituting one of the most sophisticated aspects of human intelligence [21]. This capability, primarily associated with anterior prefrontal cortex regions, enables several critical functions.

Performance Monitoring: Assessment of reasoning quality and decision accuracy during cognitive processing [22].

Confidence Calibration: Assignment of subjective confidence levels to conclusions and decisions, enabling appropriate uncertainty expression [23].

Cognitive Control: Recognition of reasoning errors, identification of knowledge gaps, and adaptive strategy modification [24].

Research demonstrates that metacognitive awareness is crucial for expert performance and prevents overconfident errors that frequently affect automated systems [25]. Individuals with superior metacognitive capabilities consistently demonstrate enhanced learning, decision-making, and performance across diverse domains.

4 The ORPA Cognitive Framework

4.1 From OODA to ORPA: Cognitive Evolution in Decision Cycles

The intellectual foundation for structured decision-making frameworks traces to Colonel John Boyd's OODA loop (Observe, Orient, Decide, Act), developed for rapid decision-making in competitive environments [26]. Boyd identified the "Orient" phase as the cognitive center of gravity, representing the complex process of sense-making that determines decision quality.

The ORPA framework represents an evolution of Boyd's insights, specifically adapted for artificial cognitive systems:

Stage	OODA	ORPA	Function
Input	Observe	Observe	Predictive processing
Cognition	Orient	Reflect	Memory integration
Decision	Decide	Plan	Future simulation
Action	Act	Act	Motor control

Table 1: Cognitive Mapping of Decision Frameworks

4.2 ORPA Stage Implementation

4.2.1 Observe: Predictive Error Detection

The Observe stage implements biological perception principles through prediction-based information processing:

Expectation Generation: Agents maintain statistical models of environmental patterns and generate specific predictions about upcoming observations based on current context and historical data.

Prediction Error Computation: Incoming data is compared against generated expectations, with significant deviations flagged as prediction errors requiring attention.

Surprise-Based Prioritization: Rather than filtering for semantic "importance," the system prioritizes statistically unexpected events that violate learned patterns, mirroring the biological mechanism that drives attention and learning.

Implementation Process: The observation processing follows a systematic approach where each new observation triggers prediction generation based on current context and historical patterns. The system then computes prediction errors by comparing actual measurements against generated expectations. When prediction errors exceed significance thresholds, the system flags these events for attention and triggers the reflection process with appropriate error context.

4.2.2 Reflect: Memory Integration and Metacognitive Evaluation

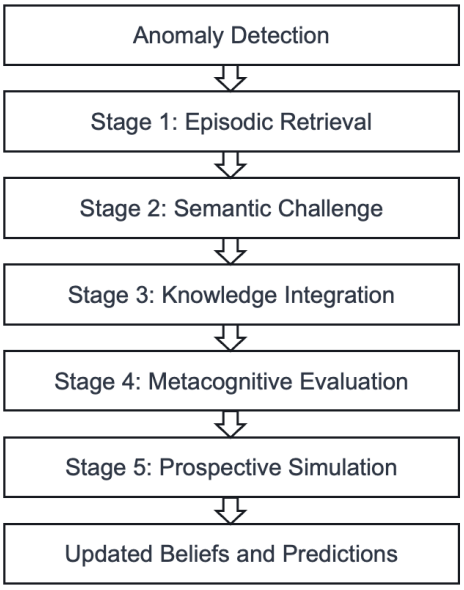


Figure 3: Multi-stage reflection process with five sequential stages: episodic retrieval, semantic model challenge, knowledge integration, metacognitive evaluation, and prospective simulation.

The Reflect stage implements sophisticated cognitive processing through multi-stage analysis:

Stage 1: Episodic Retrieval. Specific, contextually similar past experiences are retrieved from memory to provide evidentiary foundation for reasoning about current anomalies.

Stage 2: Semantic Model Challenge. Current beliefs and knowledge models are explicitly compared against new evidence, with contradictions identified and analyzed.

Stage 3: Knowledge Integration. Updated understanding is synthesized through integration of specific episodic evidence with general semantic knowledge.

Stage 4: Metacognitive Evaluation. The quality of reasoning is assessed through confidence calibration, alternative explanation generation, and uncertainty quantification.

Stage 5: Prospective Simulation. Updated knowledge is used to generate testable predictions about future scenarios, creating the cognitive bridge to planning.

4.2.3 Plan: Future Simulation and Strategy Formation

The Plan stage leverages the same neural mechanisms used for episodic memory to generate and evaluate potential future scenarios [27]. This "constructive episodic simulation" enables:

- **Scenario Generation:** Multiple potential action sequences are mentally simulated
- **Outcome Prediction:** Likely results of different strategies are evaluated
- **Risk Assessment:** Uncertainty and potential failure modes are explicitly considered
- **Strategy Selection:** Optimal approaches are chosen based on simulated outcomes

4.2.4 Act: Controlled Execution with Feedback

The Act stage executes selected plans while maintaining continuous monitoring for prediction validation and learning opportunities. This creates a closed cognitive loop where actions generate new observations that feed back into the ORPA cycle.

4.3 Enhanced Cognitive Capabilities

The ORPA framework incorporates several advanced cognitive capabilities that extend beyond basic observation and reflection:

Statistical Surprise Quantification: Rather than relying on subjective assessments of "importance," the enhanced observation system implements a normalized surprise scale (1-10) based on statistical deviation from established baselines. This quantitative approach maps directly to prediction error magnitudes: levels 1-3 represent routine variation within one standard deviation, levels 4-6 indicate notable deviations requiring monitoring, levels 7-8 represent significant anomalies demanding analysis, and levels 9-10 flag critical departures requiring immediate attention. This objective measurement system aligns with the biological attention mechanism where prediction errors drive cognitive resource allocation.

Counterfactual Reasoning Integration: Advanced implementations incorporate counterfactual analysis capabilities that examine alternative decision pathways based on historical evidence. The system analyzes "what if" scenarios by retrieving comparable situations where different approaches were taken, evaluating their outcomes, and using this comparative analysis to strengthen current decision-making. This capability mirrors the human cognitive process of learning from both direct experience and imagined alternatives, enabling agents to benefit from paths not taken as well as actions actually executed.

Multi-Agent Coordination Protocols: The framework includes sophisticated collaboration mechanisms that trigger appropriate inter-agent communication based on finding significance levels. Critical anomalies (levels 9-10) automatically alert relevant specialist agents, while cross-domain correlations flag appropriate agents for contextual analysis. This coordination approach mirrors how human expert teams naturally communicate about significant findings while avoiding information overload from routine observations.

Objective Function Integration: All cognitive processing explicitly connects insights to measurable performance objectives through weighted component analysis. This ensures that cognitive resources focus on changes that materially impact organizational goals rather than pursuing intellectually interesting but operationally irrelevant anomalies. The system maintains clear traceability between observed patterns, cognitive conclusions, and performance implications.

Anti-Hallucination Safeguards: The prompt architecture includes comprehensive data grounding requirements that mandate specific citation of source materials, prohibit fabrication of evidence not present in the provided context, and require explicit acknowledgment of uncertainty when data is insufficient. These safeguards address one of the most significant challenges in LLM-based systems by structurally preventing the generation of plausible-sounding but factually incorrect information.

4.4 Advanced Prompt Engineering for Cognitive Processes

The implementation of neuroscience-grounded cognitive processes requires sophisticated prompt engineering that guides language models through human-like reasoning patterns. The enhanced prompt architecture incorporates several critical innovations that translate cognitive science principles into practical implementation strategies.

The observation prompts implement prediction-based analysis where agents generate expectations about system behavior based on historical patterns, then identify deviations that represent genuine prediction errors rather than arbitrary importance assessments. The reflection prompts guide agents through the five-stage cognitive processing pipeline, ensuring systematic integration of episodic memories with semantic knowledge updates while maintaining metacognitive self-evaluation throughout the reasoning process.

5 ORPA as a Sequential Decision Process

Sections 3 and 4 grounded each ORPA stage in a neuroscientific finding and described the cognitive function it performs. This section reads the same four stages through a second and complementary lens, namely Powell's Sequential Decision Analytics, and asks what decision-theoretic object each stage computes. The two readings are not rivals. The neuroscientific account explains why the architecture takes the shape it does, by reference to mechanisms that decades of evidence have validated, and the decision-theoretic account explains what the shape computes and how one demonstrates that the computation is correct. The surprise signal, treated in Section 5.3, is where the two accounts turn out to be describing the same quantity.

5.1 ORPA Stages as Elements of the Universal Model

In SDA, a sequential decision problem is written with a state variable S_t that is necessary and sufficient at time t to compute the objective, execute the policy, and model the system forward; a decision variable x_t that the system controls; exogenous information W_{t+1} that arrives between decisions and is where uncertainty enters; a transition function $S_{t+1} = S^M(S_t, x_t, W_{t+1})$; and an objective defined over policies,

$$\max_{\pi} \mathbb{E}\left\{\sum_{t} C\big(S_t, X^{\pi}(S_t)\big)\right\}. \tag{1}$$

The optimization is over the policy π, that is over the method of deciding, rather than over any single decision. The four ORPA stages compute these elements in turn, and Table 2 states the correspondence.

The state itself decomposes the way Powell prescribes. The resource state holds the agent's available actions and tool library; the information state holds its memories, measures, and retrieved knowledge; and the belief state holds the probability distributions that quantify what the agent does not directly observe, which in the present architecture include both the confidence attached to each decision and, once Section 5.5 is in operation, the learned distributions over the effects of actions. Constructing the state is therefore not a passive recording step. It is the act of deciding what enters S_t, which is precisely the function that ORPA's significance and retrieval mechanisms perform.

5.2 ORPA's Policy Class, and a Deliberate Non-Goal

Read through Powell's taxonomy, the ORPA policy is a hybrid of two of the four classes. Its triggers and rules are policy function approximations: the significance threshold that initiates reflection, the deontic constraints that bound permissible actions, and the role-conditioned significance weightings are analytical functions that map state to decision with no embedded optimization. Its planning stage is a governed direct lookahead. In the Plan-and-Solve procedure the language model composes a plan over a horizon and estimates its impact, but it can only select from the actions the agent is configured and permitted to take, so a plan cannot introduce an action outside that set. Software then scores each candidate plan against the objective and checks it against the constraints before anything executes, and the agent replans when conditions change, which is the rolling-horizon pattern of model predictive control. The lookahead is therefore not a validated process-model rollout, and it is not free-form generation either: the language model proposes within a fixed action space, and deterministic scoring and validation decide which plan proceeds. This pairing of policy function approximations with a governed direct lookahead closely matches the combination that Powell documents as recurring in working industrial systems.

The architecture deliberately does not use value function approximations or the deep reinforcement learning built on them, and the omission is a design choice rather than an unfinished feature. Value-function methods dominate the research literature, but they are unstable under the combination of function approximation, bootstrapping, and off-policy updating that Sutton and Barto term the deadly triad [36], they are data-hungry, and their learned value surfaces are opaque to inspection. Industrial processes cannot supply the millions of exploratory episodes such methods require, and they cannot tolerate the exploratory failures those episodes imply. Positioning ORPA on the policy-function and lookahead branch of the theory is therefore a claim of fitness for the industrial setting, not a concession of immaturity relative to reinforcement learning agents.

5.3 The Surprise Signal as the Bridge Between the Two Groundings

The Observe stage in Section 4.2.1 assigns weight to an observation in proportion to its statistical unexpectedness, computed as a prediction error normalized by expected variance and modulated by precision. That construction was derived there from predictive coding, in which the brain minimizes surprise, formally the negative log probability of sensory input under its generative model, and in which precision is the inverse of predicted variance [7, 8, 12]. The same quantity has a second and independent grounding in information theory. Shannon defines the information content of an event as the negative logarithm of its probability, so that rare events carry more information [34]. Surprise in predictive coding and information content in Shannon's sense are the same function of the same probability, and the prediction error is the gradient signal that a system minimizes when it reduces that surprise.

ORPA stage	Decision-theoretic computation	Universal-model element
Observe	Prediction-error detection and surprise scoring decide which incoming information is admitted and with what weight.	Exogenous information W_{t+1} updating the information state; surprise gates state construction.
Reflect	Episodic retrieval, semantic-model challenge, and knowledge integration form and revise distributions over unobserved quantities; metacognitive evaluation calibrates confidence.	Belief state B_t formation and update; confidence as quantified uncertainty over the decision.
Plan	Prospective simulation generates and scores candidate action sequences against the objective, replanning as information arrives.	Direct lookahead approximation; objective-weighted policy evaluation.
Act	Controlled execution commits the chosen action and monitors the result, which becomes the next observation.	Decision x_t executed; transition S^M produces the next W.

Table 2: ORPA stages mapped to the elements of Powell's universal decision model.

This identity is the keystone of the integration, because it shows the two groundings reinforcing rather than competing. The 1 to 10 prioritization scale of Section 4.3 can be read in either vocabulary without contradiction: as the precision-weighted magnitude of a prediction error, which is the neuroscientific reading, or as a normalized quantity of information, which is the decision-theoretic reading. The decision-theoretic reading adds a consequence the neuroscientific one does not by itself supply, namely that a surprising observation is exactly an observation with high value of information, the quantity that optimal-learning theory uses to decide what to sense and when [30]. ORPA's surprise-driven attention is thus simultaneously the biological mechanism that allocates cognitive resources to the unexpected and the decision-theoretic mechanism that allocates sensing effort to where it most improves future decisions. A single signal discharges both roles, which is the strongest available evidence that the two foundations describe one architecture rather than two.

5.4 Determinism: The Executing Object Is a Function

A recurring objection to any agent that contains a language model is that it cannot be validated, because the same prompt may yield different completions, so identical conditions need not produce identical actions. SDA resolves the objection by making precise what executes. The object that executes is a policy $X^\pi(S_t \mid \theta)$, a function from a rigorously defined state to a decision, parameterized by tunable values θ. A function is deterministic by definition, so identical state and identical parameters produce an identical decision, provably. The question of determinism therefore reduces to a precise one: which components of the architecture lie inside the policy function, and which merely produce inputs to it.

In ORPA the components that bound and validate an action lie inside the policy function and belong to the deterministic policy classes. The deontic constraint rules and the significance thresholds are policy function approximations, so evaluating them against a frozen state returns the same verdict on every run. The objective and utility scoring and the plan validation are cost-function and deterministic-lookahead computations, so the same inputs and the same parameters yield the same score. The language model, by contrast, lies outside the policy function. Its output is exogenous information W, and Powell's catalog of uncertainty already contains the relevant category, namely algorithmic instability, meaning different outputs from identical runs of an algorithm. The architecture therefore does not ask the language model to be deterministic, which is impossible, but instead records its output as a noisy input and makes every downstream step a deterministic function of that record. The language model proposes, and the policy disposes.

This yields a constructive test rather than a promise. Powell defines the state as a function of history that is necessary and sufficient to compute the policy from time t onward, which, read as an engineering specification, says that if replaying the captured state does not reproduce the action, then the state capture was incomplete, by definition. The replay test is thus the proof of conformance, and it is the same discipline that already governs programmable logic controllers and model predictive control. This strengthens the reliability and anti-hallucination claims of Sections 4.3 and 7 by giving them a formal basis. The anti-hallucination safeguards prevent the cognitive layer from fabricating evidence, and the policy-function structure guarantees that whatever the cognitive layer produces is bounded by a deterministic, replayable validation layer before it can become an action.

5.5 Bounded Learning Through a Belief State

Section 4.2.2 described semanticization, the Reflect stage in which specific episodic evidence updates general semantic knowledge, as the qualitative mechanism by which an agent's beliefs change with experience. SDA gives one important case of that mechanism a quantitative and provably safe form. The case is the agent's estimate of what its actions actually do. For each pairing of an action a with

a measured outcome k, the agent maintains a Gaussian belief over the true effect,

$$B_t = \{(\bar{\mu}_{a,k}, \bar{\sigma}_{a,k}) : a \in \mathcal{A},\ k \in \mathcal{K}\}, \tag{2}$$

where $\mathcal{A}$ is the set of actions and $\mathcal{K}$ the set of outcomes, and on executing a and observing the outcome $W_{a,k}$ it applies the conjugate update

$$\bar{\mu} \leftarrow \frac{\beta\,\bar{\mu} + \beta^W W}{\beta + \beta^W}, \qquad \beta \leftarrow \beta + \beta^W, \tag{3}$$

where $\beta = 1/\bar{\sigma}^2$ is the precision of the belief and $\beta^W = 1/\sigma_W^2$ is the precision of the observation [30,31]. The initial estimate, which may be elicited from the language model, enters as the prior rather than as the answer, and operating experience moves the estimate toward the measured effect while quantifying the remaining uncertainty. The hardest part of this in practice is attribution, because plans execute several tasks at once, so a measured outcome can rarely be ascribed to a single action without confounding, and the observation precision must be modeled to absorb that confounding rather than ignored.

This learning is safe for three structural reasons, and the structure is what lets the determinism result of Section 5.4 survive the addition of learning. First, the conjugate update is closed-form arithmetic over recorded data, so the same prior, observation, and noise parameter yield the same posterior on the same implementation, which means the belief is itself replayable. Numerical equivalence across different platforms or numerical libraries is a separate property that has to be verified. Second, what learns is the belief state and the policy parameters θ, both of which live inside the policy, while the constraint set that defines the feasible region lies outside θ and has no write path from the learning process. The immutable safety boundary is therefore a type distinction rather than a procedure a reviewer must enforce, in the way that a thermostat cannot rewrite its own setpoint authority. Third, parameter changes are discrete, versioned, and approvable events, evaluated offline against cumulative objective performance on replayed history before any promotion, so the agent improves in auditable steps rather than drifting continuously. The same separation extends to sensing, where the choice of which tool to run and when is itself a decision with a value of information, and an interval-estimation rule that scores each option by its estimated value plus a tuned multiple of its uncertainty prevents an information source from being neglected merely because it was recently quiet [30, 32, 33].

5.6 Relationship to the Ten Laws of Trustworthy Autonomous Operations

The properties developed above correspond to several of the ten laws of the Digital Twin Consortium's Industrial AI Agent Manifesto, which states what trustworthy autonomous operations must guarantee without prescribing the mathematics that discharges the guarantees [35]. The determinism result of Section 5.4 discharges the law of deterministic validation and execution, since a policy is a function and the replay test is its conformance proof. The constraint set as the feasible region of the policy discharges the law of symbolic primacy, and the separation of the proposing language model from the disposing policy discharges the law of separation of control. The bounded-learning structure of Section 5.5, in which learning adjusts beliefs and parameters but has no write path to the constraint set, discharges the law of safe and secure continuous learning, and the use of cumulative objective performance as promotion evidence discharges the law of progressive autonomy. The framing is deliberately vendor-neutral, namely that sequential decision analytics supplies mathematics any conformant platform can build on. A full law-by-law treatment is beyond the scope of this paper and is developed separately.

6 Implementation: XMPro MAGS Case Study

6.1 System Architecture

The XMPro Multi-Agent Generative System (MAGS) provides a practical implementation platform for testing neuroscience-inspired cognitive architecture principles. The system architecture includes:

Agent Runtime Engine: Core processing system managing agent lifecycle and interaction

Memory Management System: Storage and retrieval mechanisms for different memory types

Prompt Engineering Framework: Structured templates implementing cognitive processing stages

Language Model Integration: Interface with large language models for natural language processing

6.2 Current State Analysis

Analysis of the XMPro MAGS implementation reveals both strengths and opportunities for cognitive enhancement:

Existing Capabilities:

- Basic ORPA cycle implementation with observation and reflection stages
- Memory storage system with importance and confidence scoring
- Prompt-based interaction framework with placeholder replacement
- Integration with external knowledge sources and real-time data

Read in the vocabulary of Section 5, these capabilities already constitute a hybrid policy of policy-function and governed-lookahead components operating over a rich information state, with confidence scoring as a partial belief

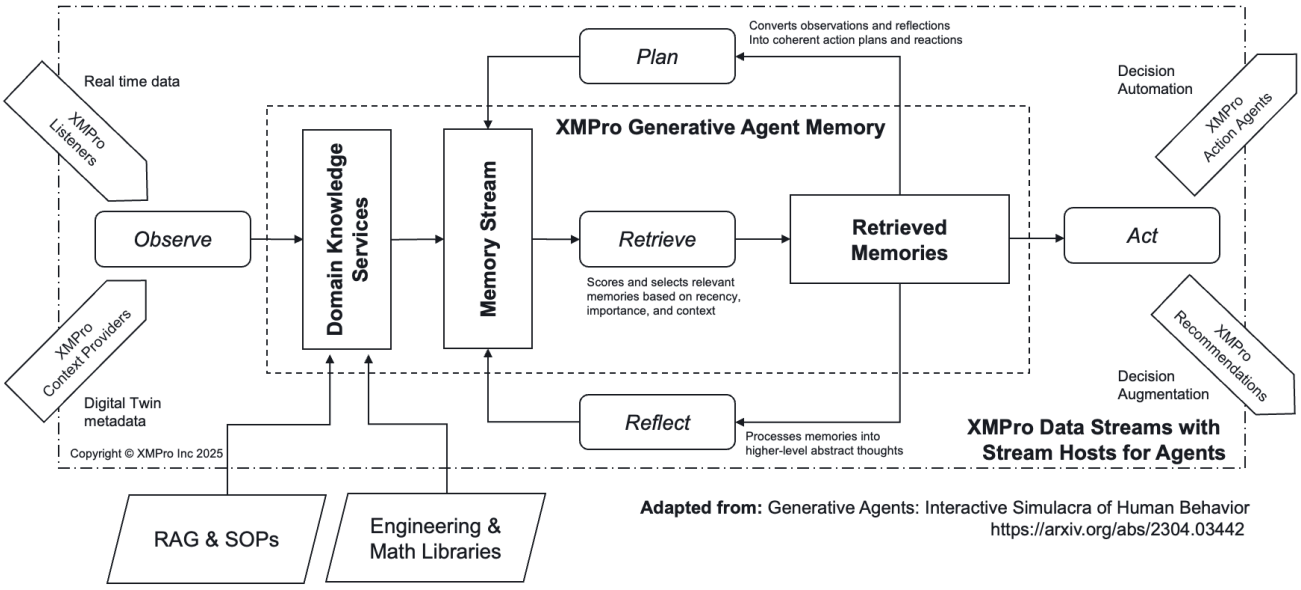


Figure 4: XMPro MAGS Cognitive Architecture implementing the ORPA cycle. The system shows real-time data flowing through the Observe stage, integration with generative agent memory systems supporting both episodic (Memory Stream) and semantic (Domain Knowledge Services) memory types, multi-stage Reflect processing, strategic Plan formation, and controlled Act execution. The architecture demonstrates practical implementation of neuroscience-inspired cognitive principles in an industrial AI platform, adapted from Stanford's Generative Agents research framework.

state; what is absent is the learned belief over action effects and the tuning of policy parameters against cumulative performance.

Enhancement Opportunities: The decision-theoretic reading of Section 5 turns the remaining gaps into named, well-studied tasks rather than open-ended research:

- An explicit belief state over the effects of actions on measures, seeded by the language model's estimate as a prior and updated from observed outcomes by the conjugate rule of Section 5.5, so that the agent becomes a better estimator with operating experience rather than regenerating point guesses each cycle.
- Treatment of the architecture's thresholds, decay factors, and weights as policy parameters θ to be tuned offline against cumulative objective performance, replacing hand-set values with values calibrated to a given deployment.
- Value-of-information selection for sensing and tool use, replacing the backward-looking interval heuristic with an interval-estimation rule that explores under-sampled sources, as in Section 5.5.
- Scenario-based plan evaluation that draws action effects from the belief distributions and reports constraint satisfaction as a frequency across sampled futures rather than a single deterministic verdict.

These are sequenced so that each is independently deployable, and each is presented here as a direction for ongoing engineering and future empirical evaluation rather than as a measured result.

6.3 Cognitive Enhancement Implementation

6.3.1 Enhanced Observation Processing

Prediction Model Integration: The enhanced observation system integrates predictive models that generate specific expectations about upcoming sensor readings based on historical patterns and current context. When new data arrives, the system computes prediction errors by comparing actual measurements against these generated expectations. An attention filtering mechanism then prioritizes events based on their surprise value rather than pre-defined importance thresholds.

The observation result includes not just the raw data, but also the magnitude of prediction errors, salient events that violate expectations, and an overall surprise score that guides subsequent cognitive processing. This approach mirrors the biological attention mechanism where unexpected stimuli automatically capture cognitive resources.

6.3.2 Multi-Stage Reflection Implementation

Cognitive Processing Pipeline: The multi-stage reflection engine implements a sequential cognitive processing pipeline that mirrors human expert reasoning patterns. The system processes the most salient observations through five distinct stages: episodic memory retrieval, semantic model challenging, knowledge integration, metacognitive evaluation, and prospective simulation.

Each stage contributes specific outputs: retrieved contextual memories, identified contradictions with existing

beliefs, updated knowledge models, confidence assessments, and simulated future scenarios. The integrated reflection result captures belief updates, calibrated confidence levels, and testable hypotheses about future outcomes. This structured approach ensures that agents engage in genuine reasoning rather than simple pattern matching.

6.4 Prompt Engineering Implementation

The prompt engineering approach translates neuroscience principles into practical language model interactions through carefully structured templates that guide cognitive processing while maintaining evidence grounding and uncertainty acknowledgment. The system implements mandatory citation requirements that prevent hallucination while enabling sophisticated reasoning about complex operational scenarios.

7 Framework Validation and Analysis

7.1 Theoretical Validation Through Neuroscience Research

The ORPA framework's effectiveness is grounded in extensive neuroscience research validating its core principles. Predictive coding theory, supported by neuroimaging studies and computational models, demonstrates that surprise-based attention allocation is fundamental to biological intelligence [8,9]. The dual memory system approach aligns with decades of research on episodic-semantic memory interactions that drive human learning and expertise [13,14].

Metacognitive research consistently shows that self-monitoring capabilities improve decision quality and prevent overconfident errors across diverse domains [21, 22]. The prospective simulation component leverages established findings about the neural overlap between remembering past events and imagining future scenarios [27].

The framework draws on a second and independent body of established theory in addition to neuroscience. Its decision-theoretic layer inherits the standing of Sequential Decision Analytics, which unifies several mature fields of stochastic optimization and control, and the determinism result of Section 5.4 is not an assertion about behavior but a constructive property with a defined test, namely that replaying the captured state must reproduce the validated action. Where the neuroscientific grounding establishes that the architecture emulates validated cognitive mechanisms, the decision-theoretic grounding establishes that what those mechanisms compute is a policy whose safety-relevant outputs are reproducible and auditable. The two groundings are mutually reinforcing rather than redundant, as Section 5.3 shows in the case of the surprise signal.

7.2 Comparison with Existing Approaches

Traditional Rule-Based Systems: Current industrial AI systems typically employ threshold-based monitoring and rule-based decision trees. These approaches excel at detecting known failure modes but struggle with novel anomalies and complex pattern recognition. They lack the adaptive learning capabilities and uncertainty quantification that characterize expert human reasoning.

Importance-Based Filtering: Many current systems attempt to prioritize events based on semantic importance rather than statistical surprise. This approach can miss subtle but significant anomalies while generating false alarms for semantically significant but statistically normal events. The prediction-error approach addresses this fundamental misalignment. In Powell's vocabulary this is the difference between scoring observations by semantic label and scoring them by the information they carry, and the prediction-error approach is the one that aligns attention with value of information.

Simple LLM-Based Agents: Basic language model implementations without cognitive architecture often suffer from hallucination issues, inconsistent reasoning, and inability to learn from experience. The structured cognitive framework addresses these limitations through evidence grounding, metacognitive evaluation, and systematic memory integration. The structural difference is that such agents place the language model in the role of the policy, whereas the architecture here places it in the role of exogenous information bounded by a deterministic policy, which is what makes its validated actions reproducible.

7.3 Implementation Insights from XMPro MAGS

The XMPro MAGS implementation provides practical insights into cognitive architecture deployment:

Cognitive Architecture Benefits: The structured ORPA cycle creates more systematic and traceable decision-making processes compared to ad-hoc prompt-based interactions. The separation of observation and reflection stages enables more focused cognitive processing and reduces the cognitive load on individual processing steps.

Evidence Grounding Effectiveness: The mandatory citation requirements and anti-hallucination safeguards significantly reduce the generation of unsupported conclusions. Agents consistently reference specific data sources and acknowledge uncertainty when evidence is insufficient.

Multi-Stage Reflection Value: The five-stage reflection process produces more comprehensive analysis than single-step summarization approaches. Each stage contributes specific cognitive functions that build toward robust conclusions and actionable insights.

Metacognitive Transparency: The confidence assessment and uncertainty quantification capabilities improve human-agent collaboration by providing calibrated assessments of conclusion reliability. This transparency builds trust and enables appropriate human oversight.

7.4 Challenges and Limitations

Computational Requirements: The multi-stage cognitive processing requires more computational resources than simple reactive systems. Organizations must balance cognitive sophistication against infrastructure costs, though improved decision quality typically justifies additional computation requirements.

Prompt Engineering Complexity: Implementing sophisticated cognitive processes requires extensive domain-specific prompt development and careful validation. The cognitive architecture benefits depend critically on prompt quality and alignment with neuroscience principles.

Integration Complexity: Cognitive architectures must integrate with existing industrial infrastructure through carefully designed abstraction layers. This integration effort can delay deployments but is necessary for practical effectiveness.

Domain Adaptation: The cognitive patterns that work effectively in one domain may require adaptation for different industrial contexts. While the underlying neuroscience principles are universal, their specific implementation needs domain expertise.

8 Discussion

8.1 Cognitive Architecture Effectiveness

The ORPA framework demonstrates that neuroscience-inspired cognitive architecture can significantly enhance multi-agent system capabilities beyond traditional approaches. The key insight is that cognitive fidelity (implementing the specific information processing patterns that characterize human expert reasoning) produces more adaptive and reliable artificial intelligence than approaches focused solely on computational power or data processing capability.

Prediction-Based Attention: The shift from importance-based filtering to prediction-error detection addresses a fundamental misalignment between traditional AI systems and biological perception mechanisms. By focusing on statistically surprising events rather than semantically significant ones, agents avoid confirmation bias patterns that often affect traditional systems.

Memory System Integration: The architectural separation of episodic experiences from semantic knowledge, with explicit mechanisms for integration during reflection, enables genuine learning rather than simple data accumulation. This approach mirrors how human experts develop expertise through experience.

Metacognitive Self-Monitoring: The addition of confidence assessment and uncertainty quantification prevents overconfident errors that frequently undermine automated system reliability. This capability is particularly crucial for building trust in human-agent collaboration scenarios.

8.2 Theoretical Implications

This work demonstrates that carefully designed prompt-based interactions can successfully implement sophisticated cognitive processes. The success of the ORPA framework suggests that cognitive architectures may be more important than underlying computational mechanisms for creating intelligent behavior, consistent with findings from the Stanford Generative Agents research [3].

The neuroscience-grounded approach provides a principled foundation for agent design that transcends domain-specific engineering. By implementing universal cognitive patterns identified through decades of neuroscience research, the framework offers a generalizable approach to creating more intelligent artificial agents.

8.3 Decision-Theoretic Implications

Reading ORPA as a sequential decision process clarifies both what the architecture claims and what it does not. The claim is that effective policies are robust, tunable, and auditable, not that they are optimal, which is consistent with Powell's own position that optimality is rarely the right target for working systems. The architecture's avoidance of value-function and deep reinforcement learning is a positioning on the practitioners' branch of the same theory, chosen because industrial processes cannot supply the data or absorb the exploratory failures those methods require. The determinism result reframes the long-standing worry about language-model variability: the variability is real and is modeled as exogenous information, and the reproducibility that validation requires is a property of the policy function that bounds it, not of the language model itself. Learning and reproducibility, often assumed to be in tension, are compatible here precisely because the learnable objects and the immutable constraint set are different kinds of object.

8.4 Practical Implementation Considerations

Computational Requirements: Implementing sophisticated cognitive processes requires more computational resources than simple workflow automation. However, modern cloud platforms and edge computing solutions make these requirements manageable, and improved decision quality typically justifies additional cost.

Integration Complexity: Cognitive architectures must work within existing industrial infrastructure. The approach addresses this through careful abstraction layer design that translates cognitive processes into standard industrial protocols while preserving the sophisticated reasoning capabilities.

Human-Agent Collaboration: The metacognitive transparency (agents expressing confidence levels and acknowledging uncertainty) proves crucial for building operator trust and enabling effective human-AI collaboration. This

capability distinguishes cognitive agents from traditional automated systems.

8.5 Limitations and Future Work

Current Limitations:

- Implementation requires extensive domain-specific prompt engineering
- Cognitive processing introduces latency compared to simple reactive systems
- Dependency on language model capabilities creates potential single points of failure

Future Research Directions:

- Integration of additional cognitive mechanisms such as working memory constraints and emotional reasoning
- Development of domain-specific cognitive patterns for different industries and expert populations
- Investigation of social cognition capabilities for enhanced multi-agent coordination
- Exploration of human-AI cognitive integration for seamless collaborative intelligence
- Controlled empirical evaluation of the bounded-learning mechanism, measuring whether action-effect beliefs converge toward observed effects under realistic confounding, and of policy-parameter tuning, measuring cumulative-objective improvement against hand-set baselines. Both are presented in this paper as directions rather than as measured outcomes.
- Joint treatment of the surprise signal as value of information, connecting the prediction-error attention of the Observe stage to the optimal-learning criterion for what to sense next.

9 Conclusion

This paper demonstrates that neuroscience-inspired cognitive architecture can transform multi-agent systems from sophisticated automation tools into genuine artificial intelligence capable of human-like reasoning, learning, and adaptation. The ORPA framework provides a practical implementation strategy that translates established cognitive science principles into functional AI systems.

Our key contributions include:

1. **Theoretical Framework:** A systematic mapping of neuroscience principles to multi-agent architecture design, grounded in predictive coding, dual memory systems, selective attention, and metacognition research.
2. **Practical Implementation:** The ORPA cognitive cycle that enables prediction-based observation, multi-stage reflection, and uncertainty-aware planning through structured prompt engineering.
3. **Enhanced Capabilities:** Advanced features including statistical surprise quantification, counterfactual reasoning, multi-agent coordination protocols, and anti-hallucination safeguards.
4. **Implementation Analysis:** Practical insights from XMPro MAGS deployment demonstrating the feasibility and benefits of neuroscience-grounded cognitive architecture.
5. **Decision-Theoretic Foundation:** A formalization of the ORPA cycle as a hybrid sequential-decision policy, under which language-model stochasticity is bounded exogenous information, the governing decision is reproducible by construction, and learning improves the agent's beliefs and parameters without ever relaxing its safety constraints.

The convergence of neuroscience research, cognitive psychology insights, and advanced language model capabilities creates unprecedented opportunities for building artificial intelligence that not only performs tasks effectively but does so through recognizably intelligent cognitive processes. The ORPA framework represents a significant step toward this goal, demonstrating that artificial agents can genuinely think, learn, and reason when built on solid cognitive science foundations.

As industrial environments become increasingly complex and dynamic, competitive advantage will belong to organizations deploying AI systems capable of expert-level cognition. This work provides both theoretical understanding and practical tools necessary to build such systems, establishing a new paradigm for multi-agent system design that prioritizes cognitive fidelity over computational sophistication.

Advances in artificial intelligence increasingly depend on understanding and implementing the cognitive architectures that characterize human intelligence rather than solely pursuing computational power or data volume expansion. Grounding AI system design in established neuroscience principles, and the decisions those agents make in established decision science, enables the development of artificial agents that demonstrate enhanced intelligence, reliability, and trustworthiness, functioning as collaborative partners in human expertise rather than simple task automation tools.

Revision Note

This revision of the September 9, 2025 paper adds a decision-theoretic layer based on Powell's Sequential Decision Analytics: Section 5 and the related-work subsection on Sequential Decision Analytics are new, and the abstract, introduction, implementation, validation, discussion and conclusion carry short additions that point to them. The neuroscientific foundations and the ORPA framework of Sections 3 and 4 are unchanged in substance.

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